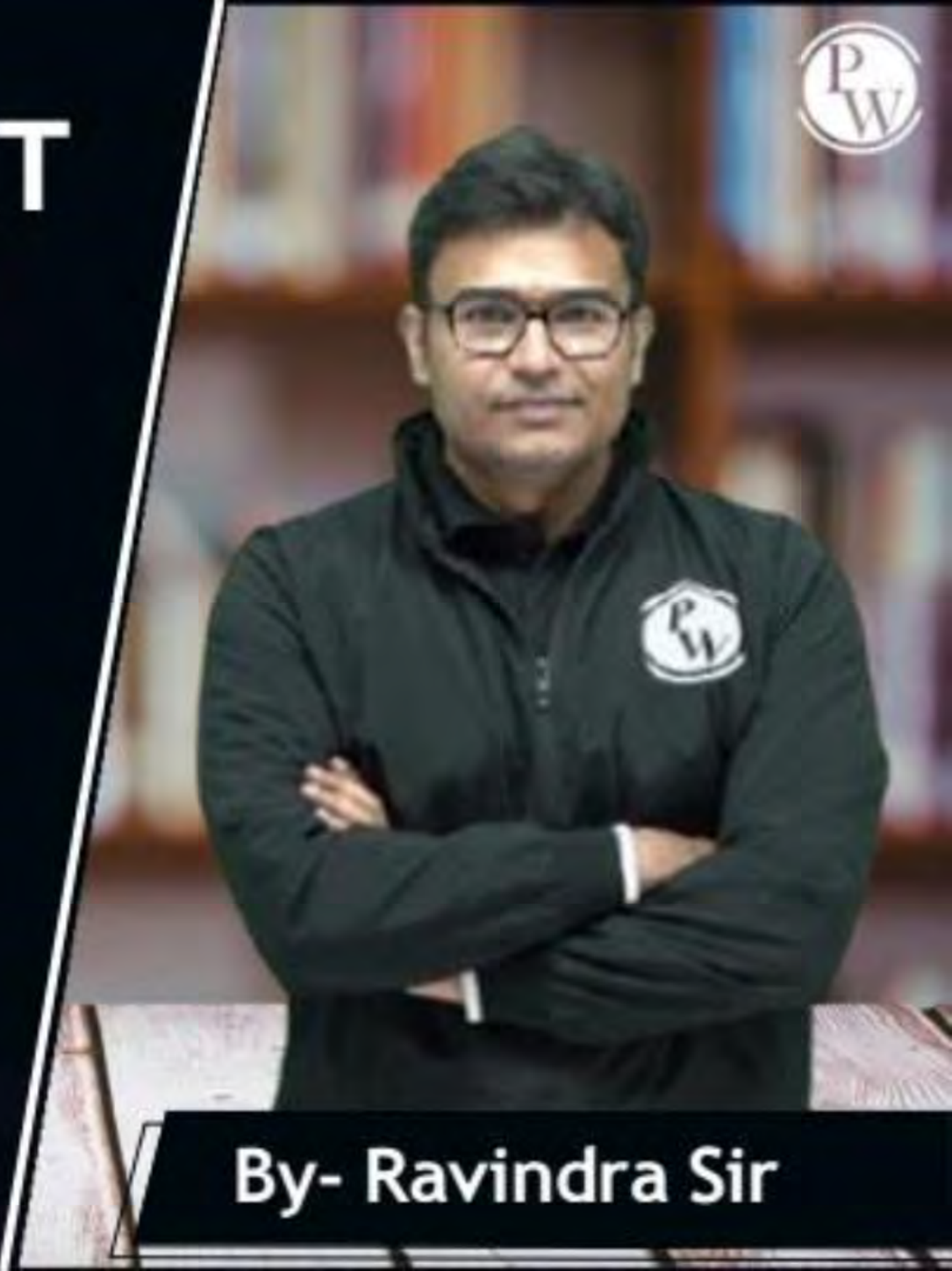
Computer Science & IT

COMPUTER NETWORK
(CN)

ISO-OSI Stack

Lecture No. 05



By- Ravindra Sir

Recap of Previous Lecture



Topic

Topic

Topic

IPV4

Topics to be Covered



- Topic
- Topic
- Topic

IPV4

Inspiring Stories : Sindhutai Sapkal



Background: Born poor in Maharashtra, faced abuse and homelessness.

Education: Studied later while doing odd jobs.

Achievements: Raised and educated over a thousand orphan children.

Impact: Known as “Mother of Orphans”; proved kindness changes lives.

Inspiring Stories : Rani Gaidinliu



Background: From a Naga village in Manipur.

Education: Village schooling.

Achievements: Led a tribal movement against British rule as a teenager.

Impact: Became a symbol of courage for Northeast India.

Inspiring Stories : Usha Mehta



Background: Born in Gujarat, loved the freedom movement.

Education: Studied law.

Achievements: Ran a secret radio during Quit India (1942).

Impact: Kept the nation's morale high when leaders were jailed.

Header checksum :-(16 bits)



Checksum is computed on the header, not on the data, why?

1) Check sum is initially calculated by sender
2) Check sum is calculated at every router.

So if we include data also in Checksum, load on the routers increases

Q) Why is checksum calculated at every router?

A) Atleast one field will change in the header at every router.

TTL will decrease at every router, so router has to reset header checksum.

→ Fields that may change in the header at the routers:

(i) Fragment offset, MF, Total length

(ii) options

(iii) Header length

TTL → Time to live

TTL will definitely change at every router

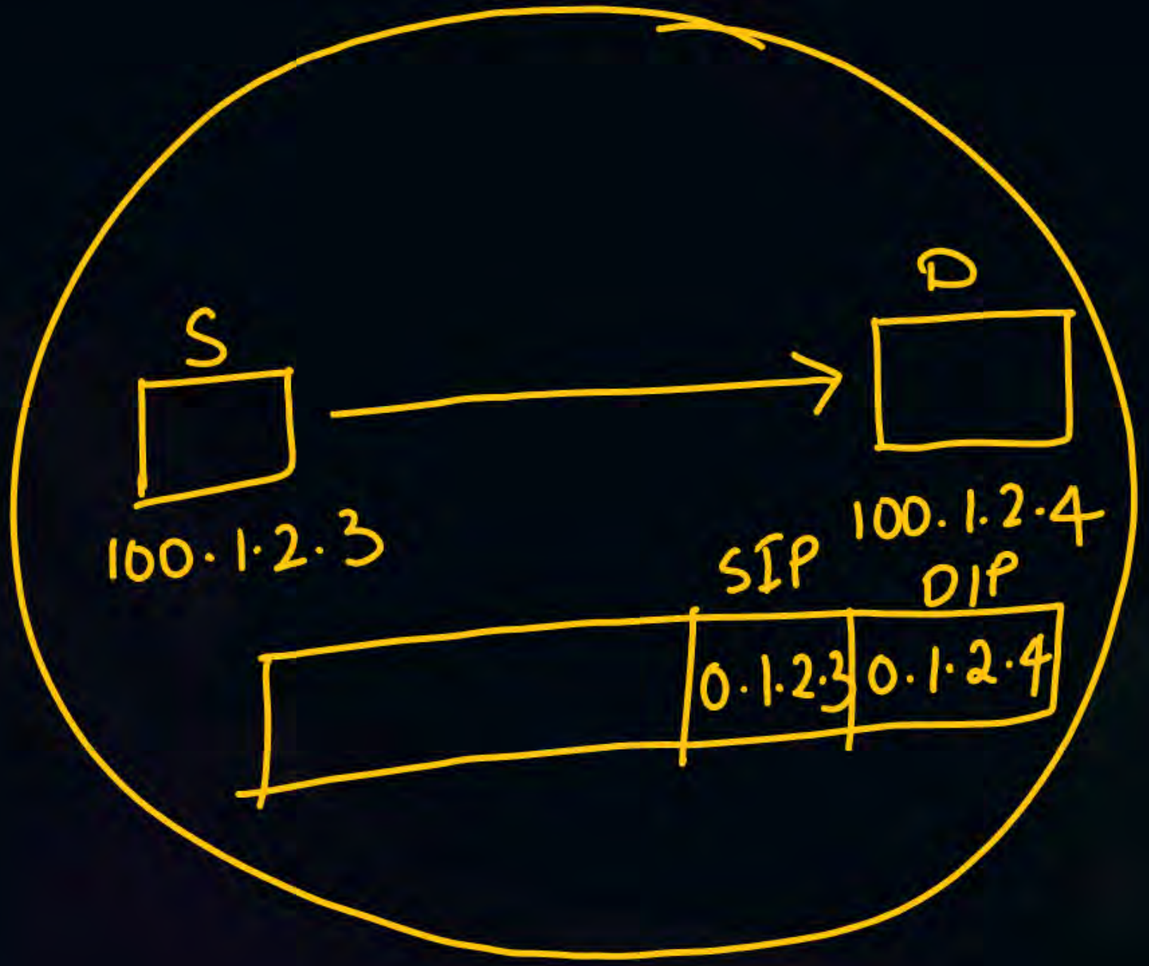
Source and destination IP add:-

	NID	HID	
SIP, DIP	✓	✓	— Valid IP
XSIP, XDIP	✓	0's	— NID
XSIP, ✓DIP	✓	1's	— DBA
XSIP, ✓DIP	1's	1's	— LBA
XSIP, XDIP	1's	0's	— SM
✓SIP, ✓DIP	0's	✓	— Host within a N/W
✓SIP, XDIP	0's	0's	— I dont have IP ✓
XSIP, ✓DIP	127	✓	— loop back add ✓

Host within a network

NID HID
0 ✓

100.0.0.0



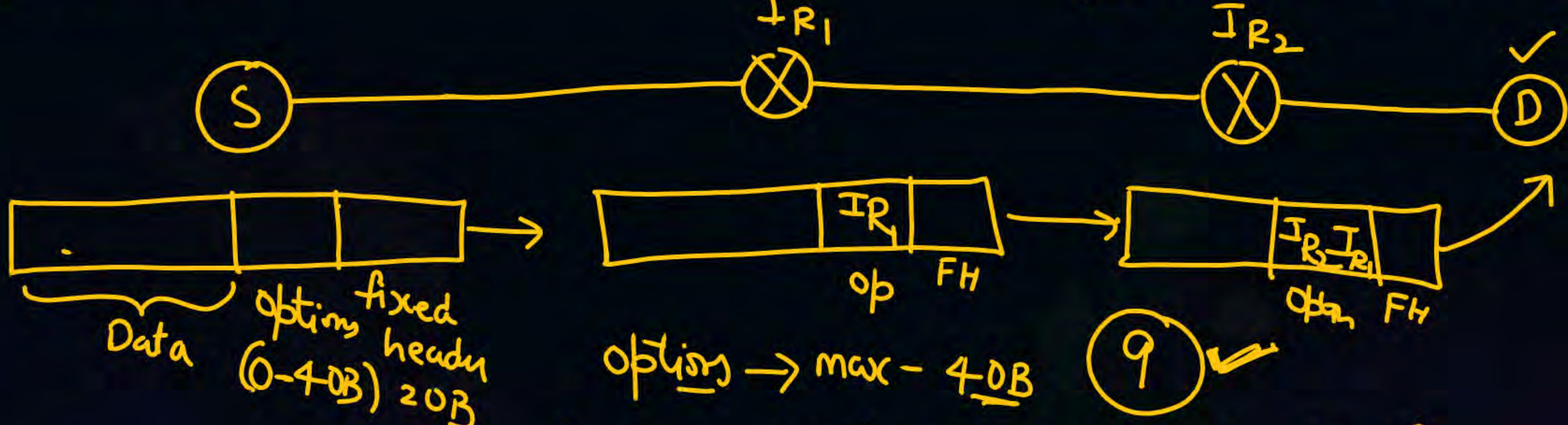
Options (0-40 Bytes)

All the datagrams need not contain options.

Options — 40B / 4B = 10 ✓

IP — Type of the option, Space

optim(i) Record Route :-

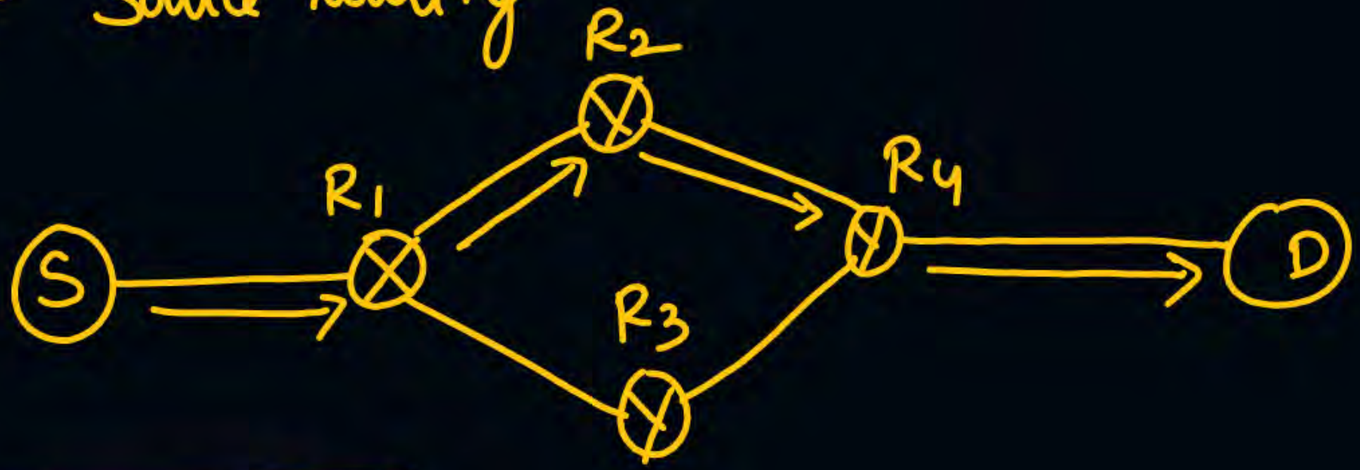


options → max - 40B

How many routers can record their address?

Ans) 9 ✓

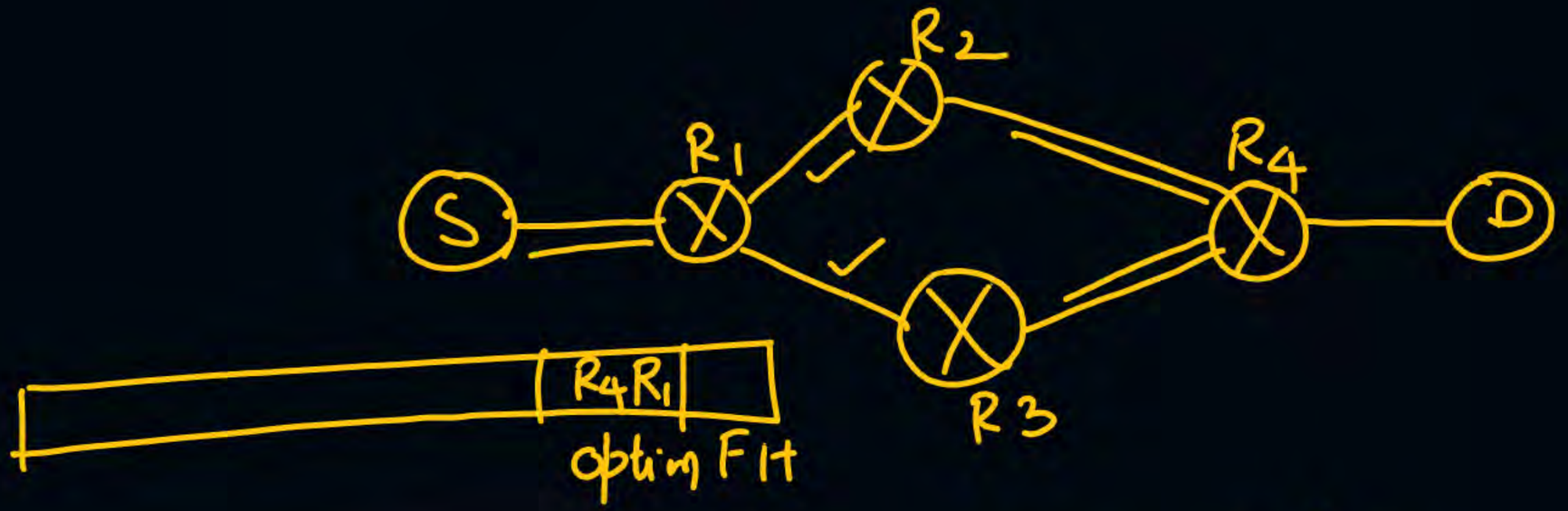
Option (ii):- Source routing





This is called strict Source routing

In strict Source routing, all the routers are specified



It is called loose source routing

In loose source routing, all the routers are not specified.

Option (iii) padding :-

If the header size is not a multiple of '4', we add dummy data to make it a multiple of 4

AL — message
TL — Segment
NL — Datagram
DLL — Frame
PL — IPDU

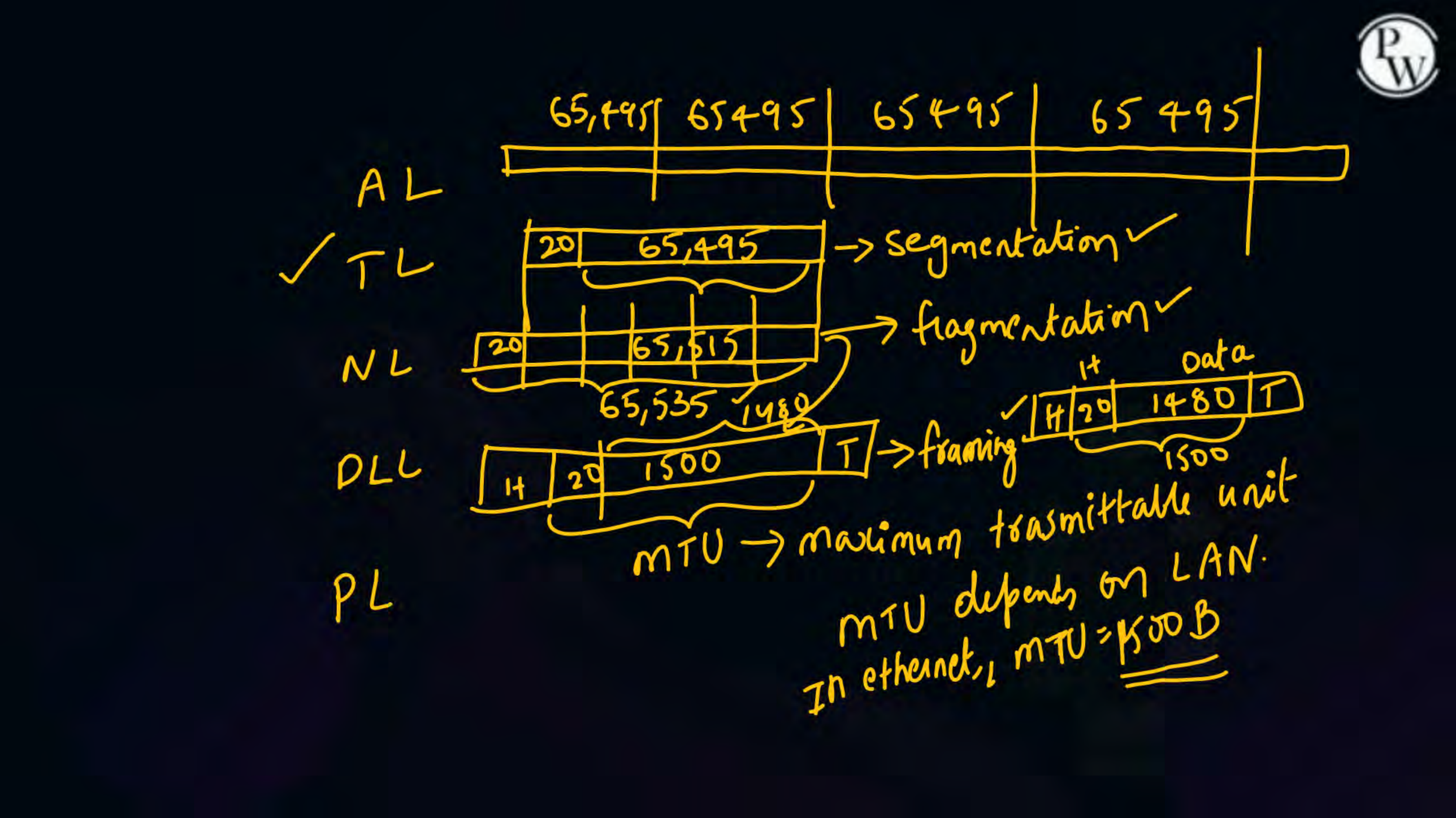
↓

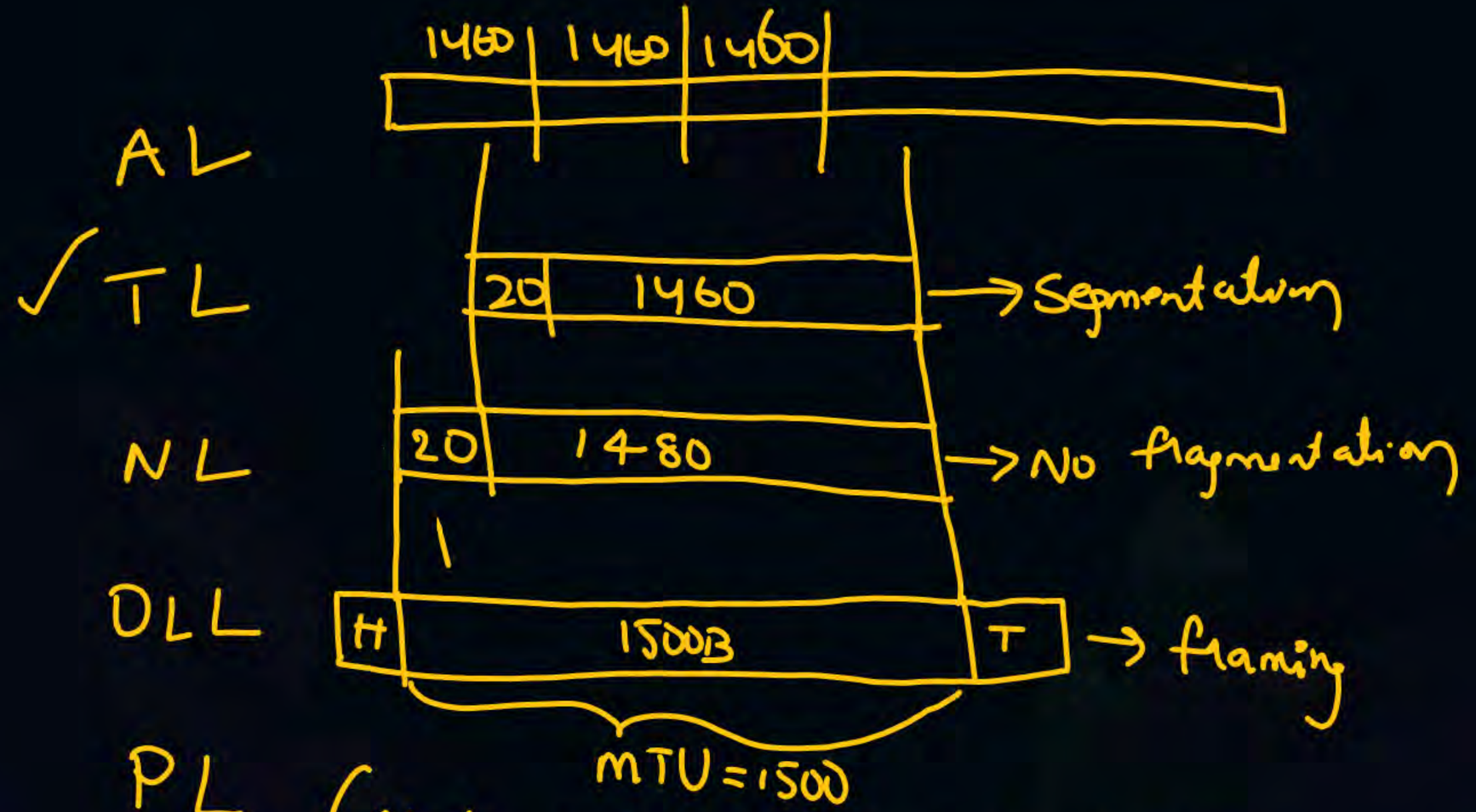
Single protocol data unit

Total length :- (16 bit)

The maximum number possible by 16 bits is $2^{16}-1$. (65,535)

So the maximum size of an IP datagram is 65535.


AL
TL
NL
DLL
PL
65,495
65495
65495
65495
20
65,495
segmentation
20
65,515
fragmentation
65,535
1480
14
20
1500
T
framing
H
Data
H
20
1480
T
1500
MTU → maximum trasmittable unit
MTU depends on LAN.
In ethernet, MTU = 1500B



If transport layer can take the bottleneck size of NL, DLL and do the Segmentation, then fragmentation is not needed

Note: At the source, we dont do flagmentation

Because TL will do the Segmentation in such a way that IP datagram fits exactly in the DLL frame.

The maximum amount of data that a DLL can carry is called MTU (maximum transmittable unit), so IP datagram max size should be equal to MTU.

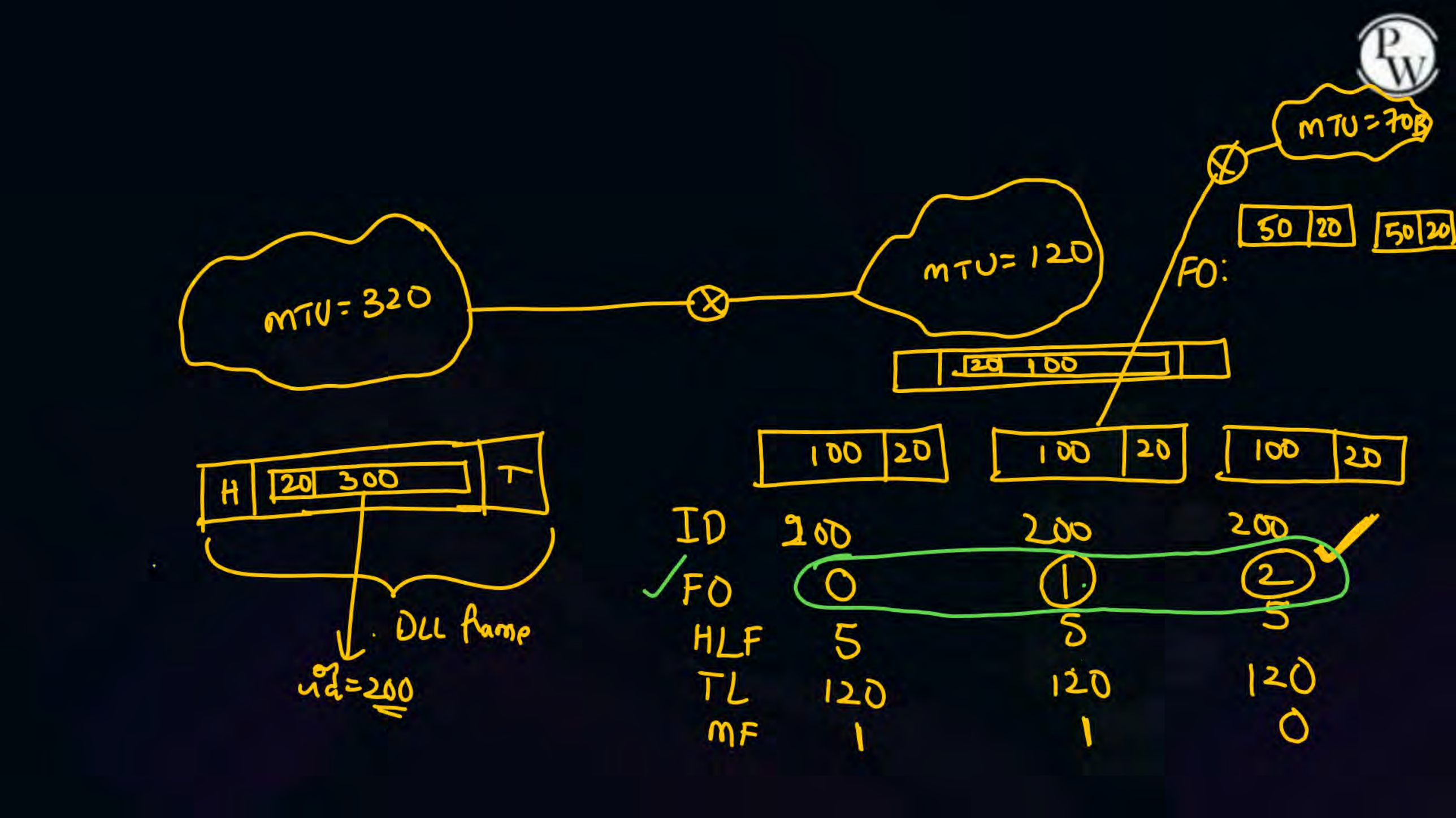
MTU varies from Lan to Lan.

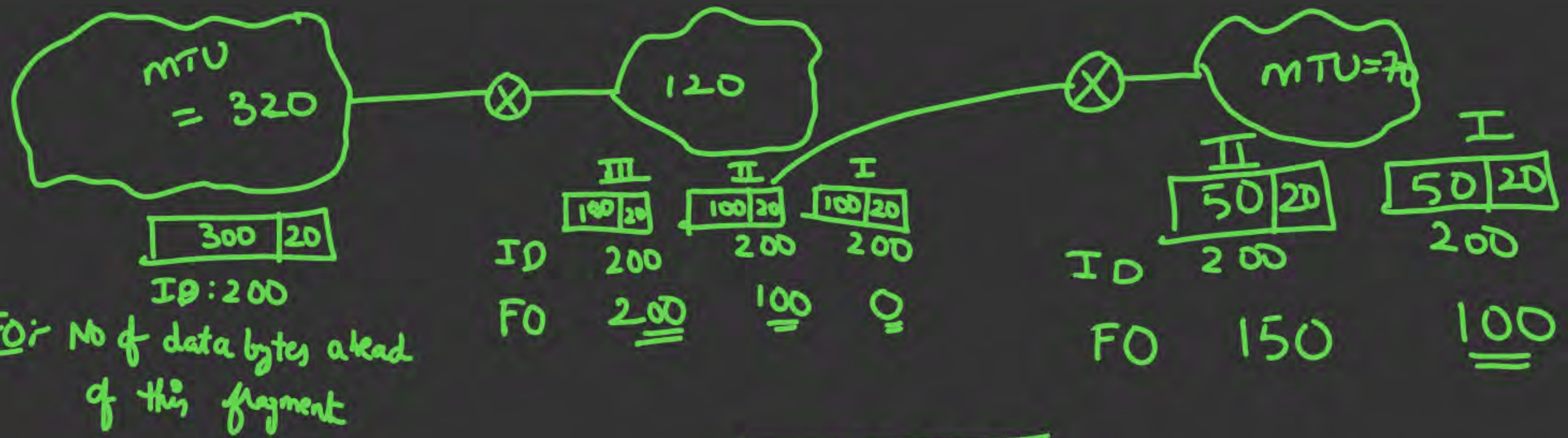


When the router has to send a DLL frame from one N/w to other N/w, fragmentation may be required if the target N/w MTU is smaller than the source N/w MTU.



Dividite a datagram into multiple fragments is called fragmentation.
All the fragment should be reassembled at the destination


MTU = 320
MTU = 120
MTU = 70B
H
20 300
T
DLL frame
पूर्व = 200
20 100
FO:
50 20
50 20
100 20
100 20
100 20
ID
FO
HLF
TL
MF
200
200
200
0
1
2
5
5
5
120
120
120
1
1
0


MTU = 320
120
MTU=7
300 20
ID: 200
FO: No of data bytes ahead of this fragment
III
II
I
100 20
100 20
100 20
ID 200 200 200
FO 200 100 0
II
I
50 20
50 20
ID 200 200
FO 150 100

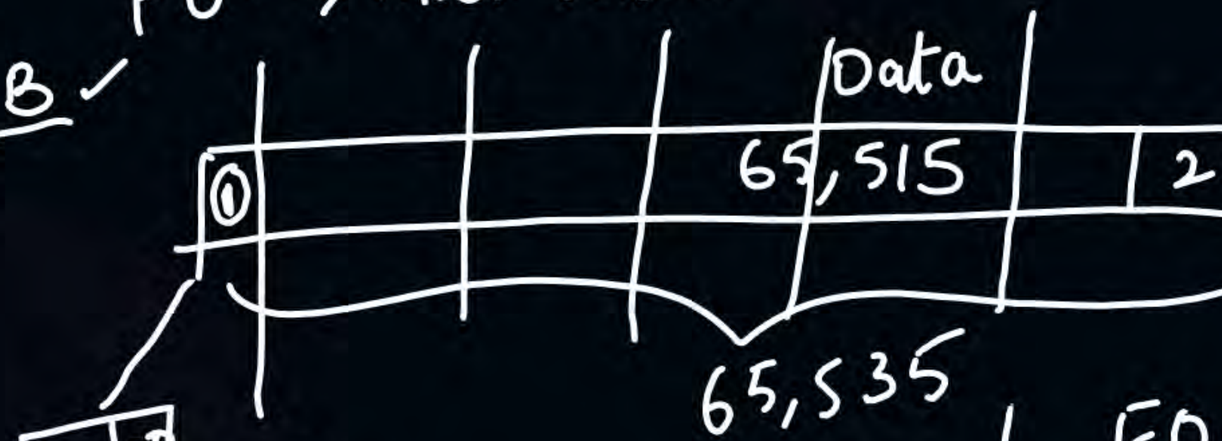

300 20
100 20
200
100
100 20
100 20
0
50 20
150
50 20
100

FO → no of data bytes ahead of this fragment

HLF = 4 bits ⇗ 15

FO → max value.

max HL = 60 B / 15 ✓



FO / 8 ⇒ FO

1 | 23

65,514 → max → FO

≃ 2^{16}

FO – 13 bits

2^{13}

Scaling factor = $\frac{2^{16}}{2^{13}} = 8$ ✓

THANK - YOU